Lecture 2 Module 2

Monday, 24 August 2020 3:39 PM

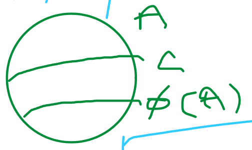
Shröder-Bernstein Theorem

Cantor, Dedekind, Shröder, Bernstein, Zermelo

If $f: A \to B$, $g: B \to A$ are both one-one maps, then there must be a bijection between A and B, i.e. $A \sim B$

e.g. $(0,1) \sim [0,1]$

Lemma: If $\phi: A \to A$ is one-one and $\phi(A) \subsetneq C \subsetneq A$ for any C, then $C \sim A$.

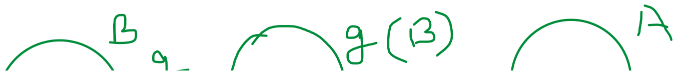


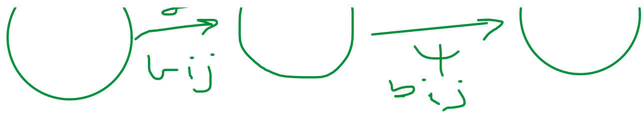
Proof of S-B theorem: Assume neither f nor g is a bijection

- $\phi := g \circ f : A \to A$ is one-one.
- Take $C := g(B)$ in Lemma.
- $g \circ f(A) \subsetneq g(B) \subsetneq A$

[Note: $f(A) \subsetneq B$, $g(B) \subsetneq A$]

- So there is $\psi : g(B) \to A$, a bijection.





<u>Proof of Lemma</u>:

Given $\phi : A \to A$ one-one; $\phi(A) \subsetneq C \subsetneq A$.

Define $X_1 := A \setminus C$.

$X_{n+1} := \phi(X_n)$, $n \in \mathbb{N}$,

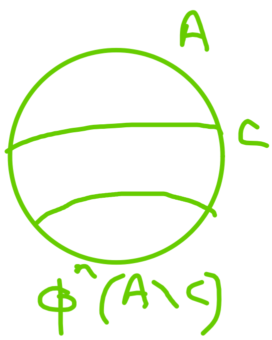
$X := \bigcup_n X_n$.

<u>Obs.</u>: $X = (A \setminus C) \cup Y$, $Y \subseteq C$.

• $A \setminus X \subseteq C$.

Define $\psi : A \to A$ as

$$\psi(a) := \begin{cases} \phi(a) & \text{if } a \in X \\ a & \text{if } a \in A \setminus X. \end{cases}$$



<u>Obs.</u> (i) $\psi : A \to C$.

• $a \in X$ ✓. • $a \in A \setminus X \Rightarrow a \notin A \setminus C$ ✓

(ii) ψ is one-one:

To show for any $a, b \in A$,

$\psi(a) = \psi(b) \Rightarrow a = b$.

• $a, b \in A \setminus X$ ✓.

• $a \in A \setminus X$, $b \in X \Rightarrow a = \psi(a)$
$= \psi(b)$
$= \underline{\phi(b) \in X}$.

• $a, b \in X$. So $\phi(a) = \psi(a) = \psi(b) = \phi(b)$

So ϕ is one-one

& ψ is one-one.

(ii) ψ is onto:

Let $b \in C$.

- $b \in A \setminus X$ ✓
- $b \in X \cap C$. $\Rightarrow b \notin X_1 \; (= A \setminus C)$

So $b = \phi(a)$, for some $a \in X$.

$= \psi(a)$. $\square$